

Plan

- Examples of optimization problems
- Examples of Linear Programming Problems (LPP)
- Solution of an LPP by the Graphical Method
- Extreme points and Corner points
- Exercises and Questions

Optimization Problems

Optimization Problems

- Nothing takes place in the world whose meaning is not that of some minimum or maximum: [L. Euler](#)

Optimization Problems

- Nothing takes place in the world whose meaning is not that of some minimum or maximum: [L. Euler](#)
- **Heron's problem:** 'On Mirrors' (book related to laws of reflection of light, approx 1st century AD).

Optimization Problems

- Nothing takes place in the world whose meaning is not that of some minimum or maximum: L. Euler
- **Heron's problem:** 'On Mirrors' (book related to laws of reflection of light, approx 1st century AD).
A and B are two given points on the same side of a line L.
Find a point D on L such that the sum of the distances from A to D and from B to D is a minimum.

Optimization Problems

- Nothing takes place in the world whose meaning is not that of some minimum or maximum: L. Euler
- Heron's problem: 'On Mirrors' (book related to laws of reflection of light, approx 1st century AD).
A and B are two given points on the same side of a line L.
Find a point D on L such that the sum of the distances from A to D and from B to D is a minimum.
- Extremal principles of nature:

Optimization Problems

- Nothing takes place in the world whose meaning is not that of some minimum or maximum: L. Euler
- **Heron's problem:** 'On Mirrors' (book related to laws of reflection of light, approx 1st century AD).
A and B are two given points on the same side of a line L. Find a point D on L such that the sum of the distances from A to D and from B to D is a minimum.
- Extremal principles of nature:
- **Laws of reflection of light.**
Nature breaks a ray of light at equal angles, if it does not unnecessarily want it to meander to no purpose.

Optimization Problems

- Nothing takes place in the world whose meaning is not that of some minimum or maximum: L. Euler
- **Heron's problem:** 'On Mirrors' (book related to laws of reflection of light, approx 1st century AD).
A and B are two given points on the same side of a line L. Find a point D on L such that the sum of the distances from A to D and from B to D is a minimum.
- **Extremal principles of nature:**
- **Laws of reflection of light.**
Nature breaks a ray of light at equal angles, if it does not unnecessarily want it to meander to no purpose.
- **Laws of refraction of light.**
What characterizes the trajectory of light moving from one point to another in a non homogeneous medium is that it is traversed in a minimum time.

- Isoperimetric problem, Oldest version: 'Aenid' of Vergil.

- Isoperimetric problem, Oldest version: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine)
Dido set off westward in search of a safe place.

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine)
Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine) Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.
Dido settled with the local leader Yarb for as much land that could be 'encircled with a bull's hide' (9th century BC).

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine) Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.
Dido settled with the local leader Yarb for as much land that could be 'encircled with a bull's hide' (9th century BC).
- **Isoperimetric problem**: Among all closed plane curves of a given length (perimeter), find the one that encloses the maximum area.

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine) Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.
Dido settled with the local leader Yarb for as much land that could be 'encircled with a bull's hide' (9th century BC).
- **Isoperimetric problem**: Among all closed plane curves of a given length (perimeter), find the one that encloses the **maximum area**. **Answer**: Circle

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine) Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.
Dido settled with the local leader Yarb for as much land that could be 'encircled with a bull's hide' (9th century BC).
- **Isoperimetric problem**: Among all closed plane curves of a given length (perimeter), find the one that encloses the **maximum area**. **Answer**: Circle
- **Isoepiphanic property of the Sphere**: The **Sphere** encloses the largest volume among all closed surfaces with the same surface area.

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine) Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.
Dido settled with the local leader Yarb for as much land that could be 'encircled with a bull's hide' (9th century BC).
- **Isoperimetric problem**: Among all closed plane curves of a given length (perimeter), find the one that encloses the **maximum area**. **Answer**: Circle
- **Isoepiphanic property of the Sphere**: The **Sphere** encloses the largest volume among all closed surfaces with the same surface area.
- **Easier problem**: Among all rectangles of a given perimeter (say 20), find the one that encloses the **maximum area**.
Maximize **ac**

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine) Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.
Dido settled with the local leader Yarb for as much land that could be 'encircled with a bull's hide' (9th century BC).
- **Isoperimetric problem**: Among all closed plane curves of a given length (perimeter), find the one that encloses the **maximum area**. **Answer**: Circle
- **Isoepiphanic property of the Sphere**: The **Sphere** encloses the largest volume among all closed surfaces with the same surface area.
- **Easier problem**: Among all rectangles of a given perimeter (say 20), find the one that encloses the **maximum area**.
Maximize **ac**
subject to, **$a + c = 10$** ,
 $a \geq 0, c \geq 0$.

- **Isoperimetric problem**, **Oldest version**: 'Aenid' of Vergil.
Escaping persecution, the Phoenician princess (Phoenicia is today's Lebanon, with parts of Syria, occupied Palestine) Dido set off westward in search of a safe place.
She liked a certain place now known as 'Bay of Tunis'.
Dido settled with the local leader Yarb for as much land that could be 'encircled with a bull's hide' (9th century BC).
- **Isoperimetric problem**: Among all closed plane curves of a given length (perimeter), find the one that encloses the **maximum area**. **Answer**: **Circle**
- **Isoepiphanic property of the Sphere**: The **Sphere** encloses the largest volume among all closed surfaces with the same surface area.
- **Easier problem**: Among all rectangles of a given perimeter (say 20), find the one that encloses the **maximum area**.
Maximize ac
subject to, $a + c = 10$,
 $a \geq 0, c \geq 0$.
Answer: Square

Linear Programming Problems

Diet Problem: For U.S soldiers, World War II

- Let there be m nutrients $N_1, N_2, \dots, N_m$ and n food products, $F_1, F_2, \dots, F_n$, available in the market which can supply these nutrients.

Linear Programming Problems

Diet Problem: For U.S soldiers, World War II

- Let there be m nutrients $N_1, N_2, \dots, N_m$ and n food products, $F_1, F_2, \dots, F_n$, available in the market which can supply these nutrients.
- For healthy survival a human being requires atleast, b_i units of the i th nutrient, $i = 1, 2, \dots, m$, respectively.

Linear Programming Problems

Diet Problem: For U.S soldiers, World War II

- Let there be m nutrients $N_1, N_2, \dots, N_m$ and n food products, $F_1, F_2, \dots, F_n$, available in the market which can supply these nutrients.
- For healthy survival a human being requires atleast, b_i units of the i th nutrient, $i = 1, 2, \dots, m$, respectively.
- Let a_{ij} be the amount of the i th nutrient (N_i) present in unit amount of the j th food product (F_j), and let $c_j, j = 1, 2, \dots, n$ be the cost of unit amount of F_j .

Linear Programming Problems

Diet Problem: For U.S soldiers, World War II

- Let there be m nutrients $N_1, N_2, \dots, N_m$ and n food products, $F_1, F_2, \dots, F_n$, available in the market which can supply these nutrients.
- For healthy survival a human being requires atleast, b_i units of the i th nutrient, $i = 1, 2, \dots, m$, respectively.
- Let a_{ij} be the amount of the i th nutrient (N_i) present in unit amount of the j th food product (F_j), and let $c_j, j = 1, 2, \dots, n$ be the cost of unit amount of F_j .
- To decide on a diet of minimum cost consisting of the n food products (in various quantities) so that one gets the required amount of each of the nutrients.

Formulation of the Diet Problem

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$
subject to
- $\sum_{j=1}^n a_{ij} x_j \geq b_i$, for $i = 1, 2, \dots, m$,

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$
subject to
- $\sum_{j=1}^n a_{ij} x_j \geq b_i$, for $i = 1, 2, \dots, m$,
 $x_j \geq 0$ for all $j = 1, 2, \dots, n$.

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$
subject to
- $\sum_{j=1}^n a_{ij} x_j \geq b_i$, for $i = 1, 2, \dots, m$,
 $x_j \geq 0$ for all $j = 1, 2, \dots, n$.
- $\mathbf{Ax} \geq \mathbf{b}$, (or alternatively as $-\mathbf{Ax} \leq -\mathbf{b}$), $\mathbf{x} \geq \mathbf{0}$,

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$
subject to
- $\sum_{j=1}^n a_{ij} x_j \geq b_i$, for $i = 1, 2, \dots, m$,
 $x_j \geq 0$ for all $j = 1, 2, \dots, n$.
- $\mathbf{Ax} \geq \mathbf{b}$, (or alternatively as $-\mathbf{Ax} \leq -\mathbf{b}$), $\mathbf{x} \geq \mathbf{0}$,
where $\mathbf{A}$ is an $m \times n$ matrix (with m rows and n columns),
where the (i, j) th entry of $\mathbf{A}$ is a_{ij} ,

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$
subject to
- $\sum_{j=1}^n a_{ij} x_j \geq b_i$, for $i = 1, 2, \dots, m$,
 $x_j \geq 0$ for all $j = 1, 2, \dots, n$.
- $\mathbf{Ax} \geq \mathbf{b}$, (or alternatively as $-\mathbf{Ax} \leq -\mathbf{b}$), $\mathbf{x} \geq \mathbf{0}$,
where $\mathbf{A}$ is an $m \times n$ matrix (with m rows and n columns),
where the (i, j) th entry of $\mathbf{A}$ is a_{ij} ,
 $\mathbf{b} = [b_1, b_2, \dots, b_m]^T$,

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$
subject to
- $\sum_{j=1}^n a_{ij} x_j \geq b_i$, for $i = 1, 2, \dots, m$,
 $x_j \geq 0$ for all $j = 1, 2, \dots, n$.
- $\mathbf{Ax} \geq \mathbf{b}$, (or alternatively as $-\mathbf{Ax} \leq -\mathbf{b}$), $\mathbf{x} \geq \mathbf{0}$,
where $\mathbf{A}$ is an $m \times n$ matrix (with m rows and n columns),
where the (i, j) th entry of $\mathbf{A}$ is a_{ij} ,
 $\mathbf{b} = [b_1, b_2, \dots, b_m]^T$, $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$,

Formulation of the Diet Problem

- Let x_j be the amount of the j th food product to be used in the diet (so $x_j \geq 0$ for $j = 1, 2, \dots, n$) then the problem reduces to (under certain simplifying assumptions):
- Min $\sum_{j=1}^n c_j x_j = \mathbf{c}^T \mathbf{x}$
subject to
- $\sum_{j=1}^n a_{ij} x_j \geq b_i$, for $i = 1, 2, \dots, m$,
 $x_j \geq 0$ for all $j = 1, 2, \dots, n$.
- $\mathbf{Ax} \geq \mathbf{b}$, (or alternatively as $-\mathbf{Ax} \leq -\mathbf{b}$), $\mathbf{x} \geq \mathbf{0}$,
where $\mathbf{A}$ is an $m \times n$ matrix (with m rows and n columns),
where the (i, j) th entry of $\mathbf{A}$ is a_{ij} ,
 $\mathbf{b} = [b_1, b_2, \dots, b_m]^T$, $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$, and $\mathbf{0}$ is the zero vector with n components.

Transportation Problem: Soviet Union 1940's

Transportation Problem: Soviet Union 1940's

- Let there be m supply stations, $S_1, S_2, \dots, S_m$ for a particular product (P) and n destination stations, $D_1, D_2, \dots, D_n$ where the product is to be transported.

Transportation Problem: Soviet Union 1940's

- Let there be m supply stations, $S_1, S_2, \dots, S_m$ for a particular product (P) and n destination stations, $D_1, D_2, \dots, D_n$ where the product is to be transported.
- Let c_{ij} be the cost of transportation of unit amount of the product (P) from S_i to D_j .

Transportation Problem: Soviet Union 1940's

- Let there be m supply stations, $S_1, S_2, \dots, S_m$ for a particular product (P) and n destination stations, $D_1, D_2, \dots, D_n$ where the product is to be transported.
- Let c_{ij} be the cost of transportation of unit amount of the product (P) from S_i to D_j .
- Let s_i be the amount of the product available at S_i and let d_j be the corresponding demand at D_j .

Transportation Problem: Soviet Union 1940's

- Let there be m supply stations, $S_1, S_2, \dots, S_m$ for a particular product (P) and n destination stations, $D_1, D_2, \dots, D_n$ where the product is to be transported.
- Let c_{ij} be the cost of transportation of unit amount of the product (P) from S_i to D_j .
- Let s_i be the amount of the product available at S_i and let d_j be the corresponding demand at D_j .
- To find x_{ij} , $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, where x_{ij} is the amount of (P) to be transported from S_i to D_j such that the demands d_j are met and the cost of transportation is minimum.

Formulation of the Transportation Problem

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- $\text{Min } \sum_{i,j} c_{ij}x_{ij}$

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- $\text{Min } \sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- Min $\sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,
- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- Min $\sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,
- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,
- $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- $\text{Min } \sum_{i,j} c_{ij} x_{ij}$

subject to

- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,

- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,

- $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

The above problem can again be written as:

- $\text{Min } \mathbf{c}^T \mathbf{x}$

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- Min $\sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,
- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,
- $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

The above problem can again be written as:

- Min $\mathbf{c}^T \mathbf{x}$
subject to $\mathbf{Ax} \leq \mathbf{b}$,

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- Min $\sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,
- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,
- $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

The above problem can again be written as:

- Min $\mathbf{c}^T \mathbf{x}$
subject to $\mathbf{Ax} \leq \mathbf{b}$,
 $\mathbf{x} \geq \mathbf{0}$,

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- Min $\sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,
- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,
- $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

The above problem can again be written as:

- Min $\mathbf{c}^T \mathbf{x}$
subject to $\mathbf{Ax} \leq \mathbf{b}$,
 $\mathbf{x} \geq \mathbf{0}$,
- where $\mathbf{A}$ is a matrix with $(m + n)$ rows and $(m \times n)$ columns,

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- Min $\sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,
- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,
- $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

The above problem can again be written as:

- Min $\mathbf{c}^T \mathbf{x}$
subject to $\mathbf{Ax} \leq \mathbf{b}$,
 $\mathbf{x} \geq \mathbf{0}$,
- where $\mathbf{A}$ is a matrix with $(m + n)$ rows and $(m \times n)$ columns,
- $\mathbf{x}, \mathbf{0}$ are vectors with $m \times n$ components

Formulation of the Transportation Problem

The problem can be modelled as (under certain simplifying assumptions)

- Min $\sum_{i,j} c_{ij}x_{ij}$
subject to
- $\sum_{j=1}^n x_{ij} \leq s_i$, for $i = 1, 2, \dots, m$,
- $\sum_{i=1}^m x_{ij} \geq d_j$, for $j = 1, 2, \dots, n$,
- $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

The above problem can again be written as:

- Min $\mathbf{c}^T \mathbf{x}$
subject to $\mathbf{Ax} \leq \mathbf{b}$,
 $\mathbf{x} \geq \mathbf{0}$,
- where $\mathbf{A}$ is a matrix with $(m+n)$ rows and $(m \times n)$ columns,
- $\mathbf{x}, \mathbf{0}$ are vectors with $m \times n$ components
 $\mathbf{b} = [s_1, \dots, s_m, -d_1, \dots, -d_n]^T$.

- The 1st row of A (the row corresponding to the first supply constraint) is $[1, 1, \dots, 1, 0, \dots, 0]^T$

- The 1st row of A (the row corresponding to the first supply constraint) is

$$[1, 1, \dots, 1, 0, \dots, 0]^T$$

1 in the first n positions and 0's elsewhere.

- The 1st row of A (the row corresponding to the first supply constraint) is
 $[1, 1, \dots, 1, 0, \dots, 0]^T$
1 in the first n positions and 0's elsewhere.
- The second row of A (the row corresponding to the second supply constraint) is
 $[0, \dots, 0, 1, 1, \dots, 1, 0, \dots, 0]^T$

- The 1st row of A (the row corresponding to the first supply constraint) is
 $[1, 1, \dots, 1, 0, \dots, 0]^T$
 1 in the first n positions and 0's elsewhere.
- The second row of A (the row corresponding to the second supply constraint) is
 $[0, \dots, 0, 1, 1, \dots, 1, 0, \dots, 0]^T$
 1 in the $(n + 1)$ th position to the $2n$ th position and 0's elsewhere.

- The 1st row of A (the row corresponding to the first supply constraint) is
 $[1, 1, \dots, 1, 0, \dots, 0]^T$
 1 in the first n positions and 0's elsewhere.
- The second row of A (the row corresponding to the second supply constraint) is
 $[0, \dots, 0, 1, 1, \dots, 1, 0, \dots, 0]^T$
 1 in the $(n + 1)$ th position to the $2n$ th position and 0's elsewhere.
- The m th row of A (the row corresponding to the m th supply constraint) is
 $[0, \dots, 0, 1, 1, \dots, 1]^T$

- The 1st row of A (the row corresponding to the first supply constraint) is
 $[1, 1, \dots, 1, 0, \dots, 0]^T$
 1 in the first n positions and 0's elsewhere.
- The second row of A (the row corresponding to the second supply constraint) is
 $[0, \dots, 0, 1, 1, \dots, 1, 0, \dots, 0]^T$
 1 in the $(n + 1)$ th position to the $2n$ th position and 0's elsewhere.
- The m th row of A (the row corresponding to the m th supply constraint) is
 $[0, \dots, 0, 1, 1, \dots, 1]^T$
 1 in the $(m - 1)n + 1$ th position to the mn th position and 0's elsewhere.

- The $(m + 1)$ th row of $\mathbf{A}$ (the row corresponding to the first destination constraint) is
 $[-1, 0, \dots, 0, -1, 0, \dots, 0, -1, 0, \dots, 0],$

- The $(m + 1)$ th row of A (the row corresponding to the first destination constraint) is
 $[-1, 0, \dots, 0, -1, 0, \dots, 0, -1, 0, \dots, 0]$,
that is -1 at the first position, $(n+1)$ th position, $(2n+1)$ th position, ..., $((m-1)n + 1)$ th position, and 0's elsewhere.

- The $(m + 1)$ th row of A (the row corresponding to the first destination constraint) is
 $[-1, 0, \dots, 0, -1, 0, \dots, 0, -1, 0, \dots, 0]$,
 that is -1 at the first position, $(n+1)$ th position, $(2n+1)$ th position, ..., $((m-1)n + 1)$ th position, and 0's elsewhere.
- The $(m + n)$ th row of A (the row corresponding to the n th (last) destination constraint) is
 $[0, \dots, -1, 0, \dots, -1, 0, \dots, -1, 0, \dots, -1, \dots, 0, \dots, -1]$,

- The $(m + 1)$ th row of A (the row corresponding to the first destination constraint) is
 $[-1, 0, \dots, 0, -1, 0, \dots, 0, -1, 0, \dots, 0]$,
 that is -1 at the first position, $(n+1)$ th position, $(2n+1)$ th position, ..., $((m-1)n + 1)$ th position, and 0's elsewhere.
- The $(m + n)$ th row of A (the row corresponding to the n th (last) destination constraint) is
 $[0, \dots, -1, 0, \dots, -1, 0, \dots, -1, 0, \dots, -1, \dots, 0, \dots, -1]$,
 that is -1 at the n th position, $2n$ th position, $3n$ th position, ..., $(m \times n)$ th position, and 0's elsewhere.

Linear Programming Problem

Given $\mathbf{c} \in \mathbb{R}^n$, a column vector with n components, $\mathbf{b} \in \mathbb{R}^m$, a column vector with m components, and an $\mathbf{A} \in \mathbb{R}^{m \times n}$, a matrix with m rows and n columns

Linear Programming Problem

Given $\mathbf{c} \in \mathbb{R}^n$, a column vector with n components, $\mathbf{b} \in \mathbb{R}^m$, a column vector with m components, and an $\mathbf{A} \in \mathbb{R}^{m \times n}$, a matrix with m rows and n columns

A **linear programming problem**(LPP) is given by :

- Max or Min $\mathbf{c}^T \mathbf{x}$

Linear Programming Problem

Given $\mathbf{c} \in \mathbb{R}^n$, a column vector with n components, $\mathbf{b} \in \mathbb{R}^m$, a column vector with m components, and an $\mathbf{A} \in \mathbb{R}^{m \times n}$, a matrix with m rows and n columns

A **linear programming problem**(LPP) is given by :

- Max or Min $\mathbf{c}^T \mathbf{x}$
- subject to $\mathbf{Ax} \leq \mathbf{b}$ (or $\mathbf{Ax} \geq \mathbf{b}$)

Linear Programming Problem

Given $\mathbf{c} \in \mathbb{R}^n$, a column vector with n components, $\mathbf{b} \in \mathbb{R}^m$, a column vector with m components, and an $\mathbf{A} \in \mathbb{R}^{m \times n}$, a matrix with m rows and n columns

A **linear programming problem**(LPP) is given by :

- Max or Min $\mathbf{c}^T \mathbf{x}$
- subject to $\mathbf{Ax} \leq \mathbf{b}$ (or $\mathbf{Ax} \geq \mathbf{b}$)
- $\mathbf{x} \geq \mathbf{0}$

Linear Programming Problem

Given $\mathbf{c} \in \mathbb{R}^n$, a column vector with n components, $\mathbf{b} \in \mathbb{R}^m$, a column vector with m components, and an $\mathbf{A} \in \mathbb{R}^{m \times n}$, a matrix with m rows and n columns

A **linear programming problem**(LPP) is given by :

- Max or Min $\mathbf{c}^T \mathbf{x}$
- subject to $\mathbf{Ax} \leq \mathbf{b}$ (or $\mathbf{Ax} \geq \mathbf{b}$)
- $\mathbf{x} \geq \mathbf{0}$
- The function $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$ is called the **objective function**, the constraints $\mathbf{x} \geq \mathbf{0}$ are called the **non negativity constraints** of the LPP.

Note that the above problem can also be written as:

- Max or Min $\mathbf{c}^T \mathbf{x}$

Note that the above problem can also be written as:

- Max or Min $\mathbf{c}^T \mathbf{x}$
- $\mathbf{a}_i^T \mathbf{x} \leq b_i$ for $i = 1, 2, \dots, m$

Note that the above problem can also be written as:

- Max or Min $\mathbf{c}^T \mathbf{x}$
- $\mathbf{a}_i^T \mathbf{x} \leq b_i$ for $i = 1, 2, \dots, m$
- $-\mathbf{e}_j^T \mathbf{x} \leq 0$ for $j = 1, 2, \dots, n$

Note that the above problem can also be written as:

- Max or Min $\mathbf{c}^T \mathbf{x}$
- $\mathbf{a}_i^T \mathbf{x} \leq b_i$ for $i = 1, 2, \dots, m$
- $-\mathbf{e}_j^T \mathbf{x} \leq 0$ for $j = 1, 2, \dots, n$

where $\mathbf{a}_i^T$ is the i th row of the matrix $\mathbf{A}$, and $\mathbf{e}_j$ is the j th column of the **Identity matrix** of order n , $\mathbf{I}_n$

Note that the above problem can also be written as:

- Max or Min $\mathbf{c}^T \mathbf{x}$
- $\mathbf{a}_i^T \mathbf{x} \leq b_i$ for $i = 1, 2, \dots, m$
- $-\mathbf{e}_j^T \mathbf{x} \leq 0$ for $j = 1, 2, \dots, n$

where $\mathbf{a}_i^T$ is the i th row of the matrix $\mathbf{A}$, and $\mathbf{e}_j$ is the j th column of the **Identity matrix** of order n , $\mathbf{I}_n$

Note that each of the functions

- $\mathbf{a}_i^T \mathbf{x}$, for $i = 1, 2, \dots, m$

Note that the above problem can also be written as:

- Max or Min $\mathbf{c}^T \mathbf{x}$
- $\mathbf{a}_i^T \mathbf{x} \leq b_i$ for $i = 1, 2, \dots, m$
- $-\mathbf{e}_j^T \mathbf{x} \leq 0$ for $j = 1, 2, \dots, n$

where $\mathbf{a}_i^T$ is the i th row of the matrix $\mathbf{A}$, and $\mathbf{e}_j$ is the j th column of the **Identity matrix** of order n , $\mathbf{I}_n$

Note that each of the functions

- $\mathbf{a}_i^T \mathbf{x}$, for $i = 1, 2, \dots, m$
- $-\mathbf{e}_j^T \mathbf{x}$, for $j = 1, 2, \dots, n$

Note that the above problem can also be written as:

- Max or Min $\mathbf{c}^T \mathbf{x}$
- $\mathbf{a}_i^T \mathbf{x} \leq b_i$ for $i = 1, 2, \dots, m$
- $-\mathbf{e}_j^T \mathbf{x} \leq 0$ for $j = 1, 2, \dots, n$

where $\mathbf{a}_i^T$ is the i th row of the matrix $\mathbf{A}$, and $\mathbf{e}_j$ is the j th column of the **Identity matrix** of order n , $\mathbf{I}_n$

Note that each of the functions

- $\mathbf{a}_i^T \mathbf{x}$, for $i = 1, 2, \dots, m$
- $-\mathbf{e}_j^T \mathbf{x}$, for $j = 1, 2, \dots, n$
- and $\mathbf{c}^T \mathbf{x}$ are all **linear functions** from $\mathbb{R}^n \rightarrow \mathbb{R}$, hence the name **linear programming problem**.

Linear function, Feasible solution, Optimal Solution

- A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a **linear map** (linear function, linear transformation) if it satisfies the following:

Linear function, Feasible solution, Optimal Solution

- A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a **linear map** (linear function, linear transformation) if it satisfies the following:

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Linear function, Feasible solution, Optimal Solution

- A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a **linear map** (linear function, linear transformation) if it satisfies the following:

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

$$\text{and } T(\alpha \mathbf{x}) = \alpha T(\mathbf{x}) \text{ for all } \alpha \in \mathbb{R} \text{ and all } \mathbf{x} \in \mathbb{R}^n$$

Linear function, Feasible solution, Optimal Solution

- A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a **linear map** (linear function, linear transformation) if it satisfies the following:

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

$$\text{and } T(\alpha \mathbf{x}) = \alpha T(\mathbf{x}) \text{ for all } \alpha \in \mathbb{R} \text{ and all } \mathbf{x} \in \mathbb{R}^n$$

- An $\mathbf{x} \geq \mathbf{0}$ satisfying the constraints $A\mathbf{x} \leq \mathbf{b}$ (or $A\mathbf{x} \geq \mathbf{b}$) is called a **feasible solution** of the linear programming problem (**LPP**)

Linear function, Feasible solution, Optimal Solution

- A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a **linear map** (linear function, linear transformation) if it satisfies the following:

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

$$\text{and } T(\alpha \mathbf{x}) = \alpha T(\mathbf{x}) \text{ for all } \alpha \in \mathbb{R} \text{ and all } \mathbf{x} \in \mathbb{R}^n$$

- An $\mathbf{x} \geq \mathbf{0}$ satisfying the constraints $A\mathbf{x} \leq \mathbf{b}$ (or $A\mathbf{x} \geq \mathbf{b}$) is called a **feasible solution** of the linear programming problem (**LPP**)
- The set of all feasible solutions of a LPP is called the **feasible region** of the LPP

Linear function, Feasible solution, Optimal Solution

- A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a **linear map** (linear function, linear transformation) if it satisfies the following:

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

$$\text{and } T(\alpha \mathbf{x}) = \alpha T(\mathbf{x}) \text{ for all } \alpha \in \mathbb{R} \text{ and all } \mathbf{x} \in \mathbb{R}^n$$

- An $\mathbf{x} \geq \mathbf{0}$ satisfying the constraints $A\mathbf{x} \leq \mathbf{b}$ (or $A\mathbf{x} \geq \mathbf{b}$) is called a **feasible solution** of the linear programming problem (**LPP**)
- The set of all feasible solutions of a LPP is called the **feasible region** of the LPP

The **feasible region** of a LPP, denoted by $\text{Fea}(\text{LPP})$ is given by,

$$\text{Fea}(\text{LPP}) = \{ \mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}, A\mathbf{x} \leq \mathbf{b} \}.$$

- Fea(LPP)

- Fea(LPP)

$$= \{\mathbf{x} \in \mathbb{R}^n : -\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, \dots, n, \quad \mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, \dots, m\}$$

- Fea(LPP)

$$= \{\mathbf{x} \in \mathbb{R}^n : -\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, \dots, n, \mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, \dots, m\}$$

where $\mathbf{a}_i^T$ is the i th row of the matrix A , and $\mathbf{e}_j$ is the j th standard unit vector, or the j th column of the identity matrix I_n

- Fea(LPP)

$$= \{\mathbf{x} \in \mathbb{R}^n : -\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, \dots, n, \quad \mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, \dots, m\}$$

where $\mathbf{a}_i^T$ is the i th row of the matrix $\mathbf{A}$, and $\mathbf{e}_j$ is the j th standard unit vector, or the j th column of the identity matrix I_n

- A **feasible solution** of an LPP is called an **optimal solution** if it **minimizes** or **maximizes** the **objective function** (depending on the nature of the problem)

- Fea(LPP)

$$= \{\mathbf{x} \in \mathbb{R}^n : -\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, \dots, n, \quad \mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, \dots, m\}$$

where $\mathbf{a}_i^T$ is the i th row of the matrix $\mathbf{A}$, and $\mathbf{e}_j$ is the j th standard unit vector, or the j th column of the identity matrix I_n

- A **feasible solution** of an LPP is called an **optimal solution** if it **minimizes** or **maximizes** the **objective function** (depending on the nature of the problem)
- If the LPP has an **optimal solution**, then the value of the objective function $\mathbf{c}^T \mathbf{x}_0$ where $\mathbf{x}_0$ is an **optimal solution** of the LPP is called the **optimal value** of the LPP.

- **Example 1:** Given the linear programming problem

Max $5x + 2y$

subject to

$$3x + 2y \leq 6$$

- **Example 1:** Given the linear programming problem

Max $5x + 2y$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1,$$

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1, -x + y \leq 1,$$

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4,$$

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0$$

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0$$

$[0, 0]^T$ is **NOT** a feasible solution

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0$$

$[0, 0]^T$ is **NOT** a feasible solution

$[1, 0]^T, [0, 1]^T$ are feasible solutions which are **NOT** optimal

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

$[3, 0]^T$ is **NOT** a feasible solution

$[0, 0]^T, [1, 0]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=10

- **Example 2:** Consider the problem,

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0$$

$[0, 0]^T$ is **NOT** a feasible solution

$[1, 0]^T, [0, 1]^T$ are feasible solutions which are **NOT** optimal

Optimal solution = $[2, 0]^T$, Optimal value=-2

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:

$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\} \text{ for some } \mathbf{a} \in \mathbb{R}^n \text{ and } d \in \mathbb{R}$$

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:
$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\}$$
 for some $\mathbf{a} \in \mathbb{R}^n$ and $d \in \mathbb{R}$
- So geometrically a **hyperplane** in $\mathbb{R}$ is an element of $\mathbb{R}$ (**a single point**),

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:

$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\} \text{ for some } \mathbf{a} \in \mathbb{R}^n \text{ and } d \in \mathbb{R}$$

- So geometrically a **hyperplane** in $\mathbb{R}$ is an element of $\mathbb{R}$ (**a single point**), in $\mathbb{R}^2$ it is a **straight line**

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:
$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\}$$
 for some $\mathbf{a} \in \mathbb{R}^n$ and $d \in \mathbb{R}$
- So geometrically a **hyperplane** in $\mathbb{R}$ is an element of $\mathbb{R}$ (**a single point**), in $\mathbb{R}^2$ it is a **straight line** in $\mathbb{R}^3$ it is the usual **plane** we are familiar with

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:

$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\} \text{ for some } \mathbf{a} \in \mathbb{R}^n \text{ and } d \in \mathbb{R}$$

- So geometrically a **hyperplane** in $\mathbb{R}$ is an element of $\mathbb{R}$ (**a single point**), in $\mathbb{R}^2$ it is a **straight line** in $\mathbb{R}^3$ it is the usual **plane** we are familiar with
- $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T (\mathbf{x} - \mathbf{x}_0) = 0\}$ for some $\mathbf{a} \in \mathbb{R}^n$, $d \in \mathbb{R}$, and $\mathbf{x}_0$ satisfying $\mathbf{a}^T \mathbf{x}_0 = d$

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:
 $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\}$ for some $\mathbf{a} \in \mathbb{R}^n$ and $d \in \mathbb{R}$
- So geometrically a **hyperplane** in $\mathbb{R}$ is an element of $\mathbb{R}$ (**a single point**), in $\mathbb{R}^2$ it is a **straight line** in $\mathbb{R}^3$ it is the usual **plane** we are familiar with
- $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T (\mathbf{x} - \mathbf{x}_0) = 0\}$ for some $\mathbf{a} \in \mathbb{R}^n$, $d \in \mathbb{R}$, and $\mathbf{x}_0$ satisfying $\mathbf{a}^T \mathbf{x}_0 = d$
- The vector $\mathbf{a}$ is called a **normal** to the hyperplane H ,

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:
 $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\}$ for some $\mathbf{a} \in \mathbb{R}^n$ and $d \in \mathbb{R}$
- So geometrically a **hyperplane** in $\mathbb{R}$ is an element of $\mathbb{R}$ (**a single point**), in $\mathbb{R}^2$ it is a **straight line** in $\mathbb{R}^3$ it is the usual **plane** we are familiar with
- $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T (\mathbf{x} - \mathbf{x}_0) = 0\}$ for some $\mathbf{a} \in \mathbb{R}^n$, $d \in \mathbb{R}$, and $\mathbf{x}_0$ satisfying $\mathbf{a}^T \mathbf{x}_0 = d$
- The vector $\mathbf{a}$ is called a **normal** to the hyperplane H , since it is **orthogonal** (or perpendicular) to each of the vectors $\mathbf{x} - \mathbf{x}_0$ on the hyperplane with tail at $\mathbf{x}_0$

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:
 $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\}$ for some $\mathbf{a} \in \mathbb{R}^n$ and $d \in \mathbb{R}$
- So geometrically a **hyperplane** in $\mathbb{R}$ is an element of $\mathbb{R}$ (**a single point**), in $\mathbb{R}^2$ it is a **straight line** in $\mathbb{R}^3$ it is the usual **plane** we are familiar with
- $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T (\mathbf{x} - \mathbf{x}_0) = 0\}$ for some $\mathbf{a} \in \mathbb{R}^n$, $d \in \mathbb{R}$, and $\mathbf{x}_0$ satisfying $\mathbf{a}^T \mathbf{x}_0 = d$
- The vector $\mathbf{a}$ is called a **normal** to the hyperplane H , since it is **orthogonal** (or perpendicular) to each of the vectors $\mathbf{x} - \mathbf{x}_0$ on the hyperplane with tail at $\mathbf{x}_0$
- All vectors $c\mathbf{a}$, $c \neq 0$ may be regarded as normals to H

- A collection of hyperplanes $H_1, \dots, H_k$ in $\mathbb{R}^n$ is **Linearly Independent (LI)** if the corresponding normal vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ are linearly independent as vectors in $\mathbb{R}^n$

- A collection of hyperplanes $H_1, \dots, H_k$ in $\mathbb{R}^n$ is **Linearly Independent (LI)** if the corresponding normal vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ are linearly independent as vectors in $\mathbb{R}^n$
- Otherwise the collection of hyperplanes is **Linearly Dependent (LD)**

- A collection of hyperplanes $H_1, \dots, H_k$ in $\mathbb{R}^n$ is **Linearly Independent (LI)** if the corresponding normal vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ are **linearly independent** as vectors in $\mathbb{R}^n$
- Otherwise the collection of hyperplanes is **Linearly Dependent (LD)**
- **Definition:** Vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ in $\mathbb{R}^n$ are **LD** if there exists real numbers $c_1, \dots, c_k$, not all zeros such that

$$c_1 \mathbf{a}_1 + \dots + c_k \mathbf{a}_k = \mathbf{O}, \quad (**)$$
 where $\mathbf{O}$ is the zero vector

- A collection of hyperplanes $H_1, \dots, H_k$ in $\mathbb{R}^n$ is **Linearly Independent (LI)** if the corresponding normal vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ are **linearly independent** as vectors in $\mathbb{R}^n$
- Otherwise the collection of hyperplanes is **Linearly Dependent (LD)**
- **Definition:** Vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ in $\mathbb{R}^n$ are **LD** if there exists real numbers $c_1, \dots, c_k$, not all zeros such that
$$c_1 \mathbf{a}_1 + \dots + c_k \mathbf{a}_k = \mathbf{O}, \quad (**)$$
 where $\mathbf{O}$ is the zero vector
- If they are not **LD** then the vectors are **LI**, then the only solution to **(**)** is $c_1 = \dots = c_k = 0$

- For $k = 2$, $\mathbf{a}_1, \mathbf{a}_2$ is **LD** $\Leftrightarrow$ there is a solution to **(**)** with either one or both c_1, c_2 nonzero
 $\mathbf{a}_1 = -\frac{c_2}{c_1}\mathbf{a}_2$ (if $c_1 \neq 0$)

- For $k = 2$, $\mathbf{a}_1, \mathbf{a}_2$ is **LD** $\Leftrightarrow$ there is a solution to **(**)** with either one or both c_1, c_2 nonzero

$$\mathbf{a}_1 = -\frac{c_2}{c_1}\mathbf{a}_2 \text{ (if } c_1 \neq 0) \quad \mathbf{a}_2 = -\frac{c_1}{c_2}\mathbf{a}_1 \text{ (if } c_2 \neq 0)$$

- For $k = 2$, $\mathbf{a}_1, \mathbf{a}_2$ is **LD** $\Leftrightarrow$ there is a solution to **(**)** with either one or both c_1, c_2 nonzero
 $\mathbf{a}_1 = -\frac{c_2}{c_1}\mathbf{a}_2$ (if $c_1 \neq 0$) $\mathbf{a}_2 = -\frac{c_1}{c_2}\mathbf{a}_1$ (if $c_2 \neq 0$)
- Any set of vectors containing the **zero vector** is **LD**

- For $k = 2$, $\mathbf{a}_1, \mathbf{a}_2$ is **LD** $\Leftrightarrow$ there is a solution to **(**)** with either one or both c_1, c_2 nonzero

$$\mathbf{a}_1 = -\frac{c_2}{c_1}\mathbf{a}_2 \text{ (if } c_1 \neq 0) \quad \mathbf{a}_2 = -\frac{c_1}{c_2}\mathbf{a}_1 \text{ (if } c_2 \neq 0)$$

- Any set of vectors containing the **zero vector** is **LD**
For example if say $\mathbf{a}_1 = \mathbf{0}$ then

$$1\mathbf{a}_1 + 0\mathbf{a}_2 \dots + 0\mathbf{a}_k = \mathbf{0},$$

- For $k = 2$, $\mathbf{a}_1, \mathbf{a}_2$ is **LD** $\Leftrightarrow$ there is a solution to **(**)** with either one or both c_1, c_2 nonzero

$$\mathbf{a}_1 = -\frac{c_2}{c_1}\mathbf{a}_2 \text{ (if } c_1 \neq 0) \quad \mathbf{a}_2 = -\frac{c_1}{c_2}\mathbf{a}_1 \text{ (if } c_2 \neq 0)$$

- Any set of vectors containing the **zero vector** is **LD**
For example if say $\mathbf{a}_1 = \mathbf{0}$ then

$$1\mathbf{a}_1 + 0\mathbf{a}_2 \dots + 0\mathbf{a}_k = \mathbf{0},$$

there is a solution to $c_1\mathbf{a}_1 + \dots + c_k\mathbf{a}_k = \mathbf{0}$, with

$c_1 = 1, c_2 = 0, \dots, c_k = 0$ not all of which are zeros.

- For $k = 2$, $\mathbf{a}_1, \mathbf{a}_2$ is **LD** $\Leftrightarrow$ there is a solution to **(**)** with either one or both c_1, c_2 nonzero

$$\mathbf{a}_1 = -\frac{c_2}{c_1}\mathbf{a}_2 \text{ (if } c_1 \neq 0) \quad \mathbf{a}_2 = -\frac{c_1}{c_2}\mathbf{a}_1 \text{ (if } c_2 \neq 0)$$

- Any set of vectors containing the **zero vector** is **LD**
For example if say $\mathbf{a}_1 = \mathbf{0}$ then

$$1\mathbf{a}_1 + 0\mathbf{a}_2 \dots + 0\mathbf{a}_k = \mathbf{0},$$

there is a solution to $c_1\mathbf{a}_1 + \dots + c_k\mathbf{a}_k = \mathbf{0}$, with $c_1 = 1, c_2 = 0, \dots, c_k = 0$ not all of which are zeros.

- A set consisting of a single non zero vector is **LI**

- **Example 2 revisited:** Consider the problem,
Min $-x + 2y$,

- **Example 2 revisited:** Consider the problem,
Min $-x + 2y$,
subject to

- **Example 2 revisited:** Consider the problem,
Min $-x + 2y$,
subject to
 $x + 2y \geq 1$,

- **Example 2 revisited:** Consider the problem,
Min $-x + 2y$,
subject to
 $x + 2y \geq 1$, $-x + y \leq 1$,

- **Example 2 revisited:** Consider the problem,

Min $-x + 2y$,

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4,$$

- **Example 2 revisited:** Consider the problem,

Min $-x + 2y$,

subject to

$$x + 2y \geq 1, \quad -x + y \leq 1, \quad 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- **Example 2 revisited:** Consider the problem,
 Min $-x + 2y$,
 subject to
 $x + 2y \geq 1$, $-x + y \leq 1$, $2x + 4y \leq 4$, $x \geq 0, y \geq 0$.
- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,

- **Example 2 revisited:** Consider the problem,
 Min $-x + 2y$,
 subject to
 $x + 2y \geq 1$, $-x + y \leq 1$, $2x + 4y \leq 4$, $x \geq 0, y \geq 0$.
- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$

- **Example 2 revisited:** Consider the problem,
 Min $-x + 2y$,
 subject to
 $x + 2y \geq 1$, $-x + y \leq 1$, $2x + 4y \leq 4$, $x \geq 0, y \geq 0$.
- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$

- **Example 2 revisited:** Consider the problem,

Min $-x + 2y$,

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$

- **Example 2 revisited:** Consider the problem,

Min $-x + 2y$,

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.

- **Example 2 revisited:** Consider the problem,

Min $-x + 2y$,

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_1, H_3\}$ is **LI**, where $H_3 = \{[x, y]^T : x + 2y = 1\}$

- **Example 2 revisited:** Consider the problem,

Min $-x + 2y$,

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_1, H_3\}$ is **LI**, where $H_3 = \{[x, y]^T : x + 2y = 1\}$
 Normals to H_3 is $\{c[1, 2]^T, c \neq 0\}$

- **Example 2 revisited:** Consider the problem,

$$\text{Min } -x + 2y,$$

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_1, H_3\}$ is **LI**, where $H_3 = \{[x, y]^T : x + 2y = 1\}$
 Normals to H_3 is $\{c[1, 2]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[1, 2]^T\}$ is **LI** for $c, d \neq 0$.

- **Example 2 revisited:** Consider the problem,

$$\text{Min } -x + 2y,$$

subject to

$$x + 2y \geq 1, -x + y \leq 1, 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_1, H_3\}$ is **LI**, where $H_3 = \{[x, y]^T : x + 2y = 1\}$
 Normals to H_3 is $\{c[1, 2]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[1, 2]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_4, H_3\}$ is **LD**

- **Example 2 revisited:** Consider the problem,

$$\text{Min } -x + 2y,$$

subject to

$$x + 2y \geq 1, \quad -x + y \leq 1, \quad 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_1, H_3\}$ is **LI**, where $H_3 = \{[x, y]^T : x + 2y = 1\}$
 Normals to H_3 is $\{c[1, 2]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[1, 2]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_4, H_3\}$ is **LD** where $H_4 = \{[x, y]^T : 2x + 4y = 4\}$

- **Example 2 revisited:** Consider the problem,

$$\text{Min } -x + 2y,$$

subject to

$$x + 2y \geq 1, \quad -x + y \leq 1, \quad 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_1, H_3\}$ is **LI**, where $H_3 = \{[x, y]^T : x + 2y = 1\}$
 Normals to H_3 is $\{c[1, 2]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[1, 2]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_4, H_3\}$ is **LD** where $H_4 = \{[x, y]^T : 2x + 4y = 4\}$
 Normals to H_4 is $\{c[2, 4]^T, c \neq 0\}$

- **Example 2 revisited:** Consider the problem,

$$\text{Min } -x + 2y,$$

subject to

$$x + 2y \geq 1, \quad -x + y \leq 1, \quad 2x + 4y \leq 4, \quad x \geq 0, y \geq 0.$$

- $\{H_1, H_2\}$ is **LI**, where $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$ Normals to H_1 is $\{c[1, 0]^T, c \neq 0\}$
 Normals to H_2 is $\{c[0, 1]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[0, 1]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_1, H_3\}$ is **LI**, where $H_3 = \{[x, y]^T : x + 2y = 1\}$
 Normals to H_3 is $\{c[1, 2]^T, c \neq 0\}$
 $\{c[1, 0]^T, d[1, 2]^T\}$ is **LI** for $c, d \neq 0$.
- $\{H_4, H_3\}$ is **LD** where $H_4 = \{[x, y]^T : 2x + 4y = 4\}$
 Normals to H_4 is $\{c[2, 4]^T, c \neq 0\}$
 $\{[1, 2]^T, [2, 4]^T\}$ is **LD** since $2[1, 2]^T - 1[2, 4]^T = \mathbf{0}$.

- What about H_1, H_2, H_3 ?

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is LD.

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is LD.
- Associated with a hyperplane H are two closed half spaces

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is **LD**.
- Associated with a hyperplane H are two **closed half spaces**
 $H_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\}$ and

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is **LD**.
- Associated with a hyperplane H are two **closed half spaces**

$$H_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\} \text{ and}$$

$$H_2 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \geq d\}.$$

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is **LD**.
- Associated with a hyperplane H are two **closed half spaces**
 $H_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\}$ and
 $H_2 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \geq d\}$.
- The **hyperplane** H , and the two **half spaces** H_1, H_2 are **closed subsets** of $\mathbb{R}^n$

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is **LD**.
- Associated with a hyperplane H are two **closed half spaces**
 $H_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\}$ and
 $H_2 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \geq d\}$.
- The **hyperplane** H , and the two **half spaces** H_1, H_2 are **closed subsets** of $\mathbb{R}^n$ (since each of these sets contains all its boundary points, the boundary points being $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{a}^T \mathbf{x} = d$).

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is **LD**.
- Associated with a hyperplane H are two **closed half spaces**

$$H_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\} \text{ and}$$

$$H_2 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \geq d\}.$$

- The **hyperplane** H , and the two **half spaces** H_1, H_2 are **closed subsets** of $\mathbb{R}^n$ (since each of these sets contains all its boundary points, the boundary points being $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{a}^T \mathbf{x} = d$).
- For the hyperplane $H = \{[x, y]^T : x + 2y = 1\}$, as well as the half spaces $H_1 = \{[x, y]^T : x + 2y \leq 1\}$, $H_2 = \{[x, y]^T : x + 2y \geq 1\}$, the boundary points are

- What about H_1, H_2, H_3 ?
- Any set of $(n + 1)$ or more hyperplanes in $\mathbb{R}^n$ is **LD**.
- Associated with a hyperplane H are two **closed half spaces**

$$H_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\} \text{ and}$$

$$H_2 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \geq d\}.$$

- The **hyperplane** H , and the two **half spaces** H_1, H_2 are **closed subsets** of $\mathbb{R}^n$ (since each of these sets contains all its boundary points, the boundary points being $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{a}^T \mathbf{x} = d$).
- For the hyperplane $H = \{[x, y]^T : x + 2y = 1\}$, as well as the half spaces $H_1 = \{[x, y]^T : x + 2y \leq 1\}$, $H_2 = \{[x, y]^T : x + 2y \geq 1\}$, the boundary points are $\{[x, y]^T : x + 2y = 1\}$.

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**
- The **feasible region** in **Example 2** is

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**
- The **feasible region** in **Example 2** is $S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**
- The **feasible region** in **Example 2** is $S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces
 $H_1 = \{[x, y]^T : x \geq 0\},$

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**
- The **feasible region** in **Example 2** is
 $S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces
 $H_1 = \{[x, y]^T : x \geq 0\}, H_2 = \{[x, y]^T : y \geq 0\},$

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**

- The **feasible region** in **Example 2** is

$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces

$$H_1 = \{[x, y]^T : x \geq 0\}, H_2 = \{[x, y]^T : y \geq 0\},$$

$$H_3 = \{[x, y]^T : x + 2y \geq 1\},$$

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**

- The **feasible region** in **Example 2** is

$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces

$$H_1 = \{[x, y]^T : x \geq 0\}, H_2 = \{[x, y]^T : y \geq 0\},$$

$$H_3 = \{[x, y]^T : x + 2y \geq 1\}, H_4 = \{[x, y]^T : 2x + 4y \leq 4\}$$

and

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**

- The **feasible region** in **Example 2** is

$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces

$$H_1 = \{[x, y]^T : x \geq 0\}, H_2 = \{[x, y]^T : y \geq 0\},$$

$$H_3 = \{[x, y]^T : x + 2y \geq 1\}, H_4 = \{[x, y]^T : 2x + 4y \leq 4\}$$

$$\text{and } H_5 = \{[x, y]^T : -x + y \leq 1\}$$

is a **polyhedral set**

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**
- The **feasible region** in **Example 2** is
 $S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces
 $H_1 = \{[x, y]^T : x \geq 0\}$, $H_2 = \{[x, y]^T : y \geq 0\}$,
 $H_3 = \{[x, y]^T : x + 2y \geq 1\}$, $H_4 = \{[x, y]^T : 2x + 4y \leq 4\}$
and $H_5 = \{[x, y]^T : -x + y \leq 1\}$
is a **polyhedral set**
- Fea(LPP) is a **polyhedral set**.

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**
- The **feasible region** in **Example 2** is $S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces
 $H_1 = \{[x, y]^T : x \geq 0\}$, $H_2 = \{[x, y]^T : y \geq 0\}$,
 $H_3 = \{[x, y]^T : x + 2y \geq 1\}$, $H_4 = \{[x, y]^T : 2x + 4y \leq 4\}$
and $H_5 = \{[x, y]^T : -x + y \leq 1\}$
is a **polyhedral set**
- $\text{Fea}(\text{LPP})$ is a **polyhedral set**.
- Since the intersection of any collection of **closed subsets** of $\mathbb{R}^n$ is again a **closed subset** of $\mathbb{R}^n$, **Fea(LPP)** is a **closed** subset of $\mathbb{R}^n$

- A set which is the intersection of a **finite number** of **closed half spaces** is called a **polyhedral set**
- The **feasible region** in **Example 2** is $S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$ which is the intersection of the closed half spaces
 $H_1 = \{[x, y]^T : x \geq 0\}$, $H_2 = \{[x, y]^T : y \geq 0\}$,
 $H_3 = \{[x, y]^T : x + 2y \geq 1\}$, $H_4 = \{[x, y]^T : 2x + 4y \leq 4\}$
and $H_5 = \{[x, y]^T : -x + y \leq 1\}$
is a **polyhedral set**
- $\text{Fea}(\text{LPP})$ is a **polyhedral set**.
- Since the intersection of any collection of **closed subsets** of $\mathbb{R}^n$ is again a **closed subset** of $\mathbb{R}^n$, **Fea(LPP)** is a **closed** subset of $\mathbb{R}^n$
- $\text{Fea}(\text{LPP})$ contains all its boundary points

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
of an LPP are called its **defining hyperplanes**.

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
 of an LPP are called its **defining hyperplanes**.
- The associated **half spaces** are
 $\mathbf{a}_i^T \mathbf{x} \leq b_i$, $i = 1, 2, \dots, m$,

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
of an LPP are called its **defining hyperplanes**.
- The associated **half spaces** are
 $\mathbf{a}_i^T \mathbf{x} \leq b_i$, $i = 1, 2, \dots, m$,
 $x_j \geq 0$, $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} \leq 0$, $j = 1, 2, \dots, n$.

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
of an LPP are called its **defining hyperplanes**.
- The associated **half spaces** are
 $\mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, 2, \dots, m$,
 $x_j \geq 0, j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, 2, \dots, n$.
- The **defining hyperplanes** in **Example 2** are
 $H_1 = \{[x, y]^T : x = 0\}$,

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
of an LPP are called its **defining hyperplanes**.
- The associated **half spaces** are
 $\mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, 2, \dots, m$,
 $x_j \geq 0, j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, 2, \dots, n$.
- The **defining hyperplanes** in **Example 2** are
 $H_1 = \{[x, y]^T : x = 0\}, H_2 = \{[x, y]^T : y = 0\},$

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
of an LPP are called its **defining hyperplanes**.
- The associated **half spaces** are
 $\mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, 2, \dots, m$,
 $x_j \geq 0, j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, 2, \dots, n$.
- The **defining hyperplanes** in **Example 2** are
 $H_1 = \{[x, y]^T : x = 0\}$, $H_2 = \{[x, y]^T : y = 0\}$,
 $H_3 = \{[x, y]^T : x + 2y = 1\}$,

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
of an LPP are called its **defining hyperplanes**.
- The associated **half spaces** are
 $\mathbf{a}_i^T \mathbf{x} \leq b_i$, $i = 1, 2, \dots, m$,
 $x_j \geq 0$, $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} \leq 0$, $j = 1, 2, \dots, n$.
- The **defining hyperplanes** in **Example 2** are
 $H_1 = \{[x, y]^T : x = 0\}$, $H_2 = \{[x, y]^T : y = 0\}$,
 $H_3 = \{[x, y]^T : x + 2y = 1\}$, $H_4 = \{[x, y]^T : 2x + 4y = 4\}$
and

- The hyperplanes $\mathbf{a}_i^T \mathbf{x} = b_i$ for $i = 1, 2, \dots, m$,
 $x_j = 0$ for $j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} = 0$ for $j = 1, 2, \dots, n$,
of an LPP are called its **defining hyperplanes**.
- The associated **half spaces** are
 $\mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, 2, \dots, m$,
 $x_j \geq 0, j = 1, 2, \dots, n$, or $-\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, 2, \dots, n$.
- The **defining hyperplanes** in **Example 2** are
 $H_1 = \{[x, y]^T : x = 0\}$, $H_2 = \{[x, y]^T : y = 0\}$,
 $H_3 = \{[x, y]^T : x + 2y = 1\}$, $H_4 = \{[x, y]^T : 2x + 4y = 4\}$
and $H_5 = \{[x, y]^T : -x + y = 1\}$

Solution by graphical method of LPP's in two variables

- **Example 1:** Given the linear programming problem

Max $5x + 2y$

subject to

$$3x + 2y \leq 6$$

Solution by graphical method of LPP's in two variables

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

Solution by graphical method of LPP's in two variables

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

Solution by graphical method of LPP's in two variables

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

The **optimal solution** is $x = 2, y = 0$ that is $[2, 0]^T$ which is a **corner point** of the feasible region.

Solution by graphical method of LPP's in two variables

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

The **optimal solution** is $x = 2, y = 0$ that is $[2, 0]^T$ which is a **corner point** of the feasible region.

Later we will again convince ourselves (by using more rigor) that this is indeed the **optimal solution**

Solution by graphical method of LPP's in two variables

- **Example 1:** Given the linear programming problem

$$\text{Max } 5x + 2y$$

subject to

$$3x + 2y \leq 6$$

$$x + 2y \leq 4$$

$$x \geq 0, y \geq 0$$

The **optimal solution** is $x = 2, y = 0$ that is $[2, 0]^T$ which is a **corner point** of the feasible region.

Later we will again convince ourselves (by using more rigor) that this is indeed the **optimal solution**

- The **optimal value** is **10**

- **Example 2:** Consider the problem
Min $-x + 2y$
subject to

- **Example 2:** Consider the problem

Min $-x + 2y$

subject to

$$x + 2y \geq 1$$

- **Example 2:** Consider the problem

Min $-x + 2y$

subject to

$$x + 2y \geq 1$$

$$-x + y \leq 1$$

- **Example 2:** Consider the problem

Min $-x + 2y$

subject to

$$x + 2y \geq 1$$

$$-x + y \leq 1$$

$$x \geq 0, y \geq 0$$

- **Example 2:** Consider the problem

Min $-x + 2y$

subject to

$$x + 2y \geq 1$$

$$-x + y \leq 1$$

$$x \geq 0, y \geq 0$$

The above linear programming problem **does not** have an optimal solution.

- **Example 2:** Consider the problem

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1$$

$$-x + y \leq 1$$

$$x \geq 0, y \geq 0$$

The above linear programming problem **does not** have an optimal solution.

- **Example 3:** In the above problem keeping the feasible region same, if we just change the objective function to Min $2x + y$, then the new problem has a **unique** optimal solution.

- **Example 2:** Consider the problem

$$\text{Min } -x + 2y$$

subject to

$$x + 2y \geq 1$$

$$-x + y \leq 1$$

$$x \geq 0, y \geq 0$$

The above linear programming problem **does not** have an optimal solution.

- **Example 3:** In the above problem keeping the feasible region same, if we just change the objective function to Min $2x + y$, then the new problem has a **unique** optimal solution.

- **Example 4:** In **Example 2** if we similarly just change the objective function to $\text{Min } x + 2y$ then the new problem has **infinitely many** optimal solutions, although the set of optimal solutions is **bounded**

- **Example 4:** In **Example 2** if we similarly just change the objective function to $\text{Min } x + 2y$ then the new problem has **infinitely many** optimal solutions, although the set of optimal solutions is **bounded**
- A set $S \subset \mathbb{R}^n$ is said to be **bounded** if there exists an $M > 0$ such that $\|\mathbf{x}\| \leq M$ for all $\mathbf{x} \in S$

- **Example 4:** In **Example 2** if we similarly just change the objective function to $\text{Min } x + 2y$ then the new problem has **infinitely many** optimal solutions, although the set of optimal solutions is **bounded**
- A set $S \subset \mathbb{R}^n$ is said to be **bounded** if there exists an $M > 0$ such that $\|\mathbf{x}\| \leq M$ for all $\mathbf{x} \in S$
- **Example 5:** In **Example 2** if we just change the objective function to $\text{Min } y$, then the new problem has **infinitely many** optimal solutions and the set of optimal solutions is **unbounded**

- **Example 4:** In **Example 2** if we similarly just change the objective function to $\text{Min } x + 2y$ then the new problem has **infinitely many** optimal solutions, although the set of optimal solutions is **bounded**
- A set $S \subset \mathbb{R}^n$ is said to be **bounded** if there exists an $M > 0$ such that $\|\mathbf{x}\| \leq M$ for all $\mathbf{x} \in S$
- **Example 5:** In **Example 2** if we just change the objective function to $\text{Min } y$, then the new problem has **infinitely many** optimal solutions and the set of optimal solutions is **unbounded**
- **Example 6:** Max $-x + 2y$
subject to
 $x + 2y \leq 1$

- **Example 4:** In **Example 2** if we similarly just change the objective function to $\text{Min } x + 2y$ then the new problem has **infinitely many** optimal solutions, although the set of optimal solutions is **bounded**
- A set $S \subset \mathbb{R}^n$ is said to be **bounded** if there exists an $M > 0$ such that $\|\mathbf{x}\| \leq M$ for all $\mathbf{x} \in S$
- **Example 5:** In **Example 2** if we just change the objective function to $\text{Min } y$, then the new problem has **infinitely many** optimal solutions and the set of optimal solutions is **unbounded**
- **Example 6:** Max $-x + 2y$
subject to
 $x + 2y \leq 1$
 $-x + y \geq 1$

- **Example 4:** In **Example 2** if we similarly just change the objective function to $\text{Min } x + 2y$ then the new problem has **infinitely many** optimal solutions, although the set of optimal solutions is **bounded**
- A set $S \subset \mathbb{R}^n$ is said to be **bounded** if there exists an $M > 0$ such that $\|\mathbf{x}\| \leq M$ for all $\mathbf{x} \in S$
- **Example 5:** In **Example 2** if we just change the objective function to $\text{Min } y$, then the new problem has **infinitely many** optimal solutions and the set of optimal solutions is **unbounded**
- **Example 6:** Max $-x + 2y$
subject to
 $x + 2y \leq 1$
 $-x + y \geq 1$
 $x \geq 0, y \geq 0$

- **Example 4:** In **Example 2** if we similarly just change the objective function to $\text{Min } x + 2y$ then the new problem has **infinitely many** optimal solutions, although the set of optimal solutions is **bounded**
- A set $S \subset \mathbb{R}^n$ is said to be **bounded** if there exists an $M > 0$ such that $\|\mathbf{x}\| \leq M$ for all $\mathbf{x} \in S$
- **Example 5:** In **Example 2** if we just change the objective function to $\text{Min } y$, then the new problem has **infinitely many** optimal solutions and the set of optimal solutions is **unbounded**
- **Example 6:** Max $-x + 2y$
subject to
 $x + 2y \leq 1$
 $-x + y \geq 1$
 $x \geq 0, y \geq 0$
- The feasible region of the above problem is **empty**. This problem is called **infeasible**, and since it **does not** have a **feasible solution** it obviously **does not** have an **optimal solution**

- Question 1: Can there be exactly 2, 5, or say exactly 100 optimal solutions of an LPP?

- Question 1: Can there be exactly 2, 5, or say exactly 100 optimal solutions of an LPP?
- If we have two optimal solutions of an LPP then what about points in between and on the line segment joining these two solutions?

- Question 1: Can there be exactly 2, 5, or say exactly 100 optimal solutions of an LPP?
- If we have two optimal solutions of an LPP then what about points in between and on the line segment joining these two solutions?
- The set of optimal solutions of an LPP is a convex set

- Question 1: Can there be exactly 2, 5, or say exactly 100 optimal solutions of an LPP?
- If we have two optimal solutions of an LPP then what about points in between and on the line segment joining these two solutions?
- The set of optimal solutions of an LPP is a convex set
That is, if $\mathbf{x}_1$ and $\mathbf{x}_2$ are two optimal solutions of an LPP,

- Question 1: Can there be exactly 2, 5, or say exactly 100 optimal solutions of an LPP?
- If we have two optimal solutions of an LPP then what about points in between and on the line segment joining these two solutions?
- The set of optimal solutions of an LPP is a convex set
That is, if $\mathbf{x}_1$ and $\mathbf{x}_2$ are two optimal solutions of an LPP, then all $\mathbf{y}$ such that $\mathbf{y} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$, are also optimal solutions of the LPP

- A nonempty set, $S \subseteq \mathbb{R}^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$

- A nonempty set, $S \subseteq R^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$
- $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$ is called a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$

- A nonempty set, $S \subseteq R^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$
- $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$ is called a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- If in the above expression, $0 < \lambda < 1$, then the **convex combination** is said to be a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$

- A nonempty set, $S \subseteq R^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$
- $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$ is called a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- If in the above expression, $0 < \lambda < 1$, then the **convex combination** is said to be a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- **Example:** $H = \{[x, y]^T : x + 2y \leq 1\}$, is a convex set

- A nonempty set, $S \subseteq R^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$
- $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$ is called a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- If in the above expression, $0 < \lambda < 1$, then the **convex combination** is said to be a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- **Example:** $H = \{[x, y]^T : x + 2y \leq 1\}$, is a convex set
 $\mathbf{x}_1 = [0, 0]^T$, $\mathbf{x}_2 = [1, 0]^T \in H$.

- A nonempty set, $S \subseteq R^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$
- $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$ is called a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- If in the above expression, $0 < \lambda < 1$, then the **convex combination** is said to be a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- **Example:** $H = \{[x, y]^T : x + 2y \leq 1\}$, is a convex set
 $\mathbf{x}_1 = [0, 0]^T$, $\mathbf{x}_2 = [1, 0]^T \in H$.
 $[0, 0]^T = 1[0, 0]^T + 0[1, 0]^T$ is a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$,

- A nonempty set, $S \subseteq R^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$
- $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$ is called a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- If in the above expression, $0 < \lambda < 1$, then the **convex combination** is said to be a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- **Example:** $H = \{[x, y]^T : x + 2y \leq 1\}$, is a convex set
 $\mathbf{x}_1 = [0, 0]^T$, $\mathbf{x}_2 = [1, 0]^T \in H$.
 $[0, 0]^T = 1[0, 0]^T + 0[1, 0]^T$ is a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$, but **not** a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$

- A nonempty set, $S \subseteq R^n$ is said to be a **convex set** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, for all $0 \leq \lambda \leq 1$
- $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$ is called a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- If in the above expression, $0 < \lambda < 1$, then the **convex combination** is said to be a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
- **Example:** $H = \{[x, y]^T : x + 2y \leq 1\}$, is a convex set
 $\mathbf{x}_1 = [0, 0]^T$, $\mathbf{x}_2 = [1, 0]^T \in H$.
 $[0, 0]^T = 1[0, 0]^T + 0[1, 0]^T$ is a **convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$, but **not** a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$
 $[\frac{2}{3}, 0]^T = \frac{1}{3}[0, 0]^T + \frac{2}{3}[1, 0]^T$ is a **convex combination** as well as a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$

- If an LPP has **more than one optimal solution** then it has **infinitely many optimal solutions**, so the answer to **Question 1** would be a **NO**

- If an LPP has **more than one optimal solution** then it has **infinitely many optimal solutions**, so the answer to **Question 1** would be a **NO**
- **Question 2:** If the **feasible region** of a LPP is a **nonempty, bounded set** then does the LPP **always** have an **optimal solution**?

- If an LPP has **more than one optimal solution** then it has **infinitely many optimal solutions**, so the answer to **Question 1** would be a **NO**
- **Question 2:** If the **feasible region** of a LPP is a **nonempty, bounded set** then does the LPP **always** have an **optimal solution**?

The answer to this question is **YES**, due to **Weierstrass**, called the **Extreme Value Theorem**

- **Extreme Value Theorem:** If S is a nonempty, closed, bounded subset of $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ is continuous, then f attains both its minimum and maximum value in S

- **Extreme Value Theorem:** If S is a nonempty, closed, bounded subset of $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ is continuous, then f attains both its minimum and maximum value in S
- **Question 3:** Whenever an LPP has an optimal solution does there always exist **at least one** corner point which is optimal?

- **Extreme Value Theorem:** If S is a nonempty, closed, bounded subset of $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ is continuous, then f attains both its minimum and maximum value in S
- **Question 3:** Whenever an LPP has an optimal solution does there always exist **at least one** corner point which is optimal?
- Given an LPP with a nonempty feasible region, $Fea(LPP) = S \subset \mathbb{R}^n$, an $\mathbf{x} \in S$ is called a corner point of S , if $\mathbf{x}$ lies at the point of intersection of n linearly independent defining hyperplanes of S

- Consider the feasible region of **Example 2**

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$,

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$,

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$, $H_3 = \{[x, y]^T : x + 2y = 1\}$,

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$, $H_3 = \{[x, y]^T : x + 2y = 1\}$,
 $H_4 = \{[x, y]^T : 2x + 4y = 4\}$ and

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$, $H_3 = \{[x, y]^T : x + 2y = 1\}$,
 $H_4 = \{[x, y]^T : 2x + 4y = 4\}$ and
 $H_5 = \{[x, y]^T : -x + y = 1\}$

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$, $H_3 = \{[x, y]^T : x + 2y = 1\}$,
 $H_4 = \{[x, y]^T : 2x + 4y = 4\}$ and
 $H_5 = \{[x, y]^T : -x + y = 1\}$
- $[1, 0]^T, [0, 1]^T, [0, \frac{1}{2}]^T, [2, 0]^T$ are the **corner points** of S .
- $[1, 0]^T$ lies on H_2, H_3 , which are **LI**

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$,
 $H_2 = \{[x, y]^T : y = 0\}$, $H_3 = \{[x, y]^T : x + 2y = 1\}$,
 $H_4 = \{[x, y]^T : 2x + 4y = 4\}$ and
 $H_5 = \{[x, y]^T : -x + y = 1\}$
- $[1, 0]^T, [0, 1]^T, [0, \frac{1}{2}]^T, [2, 0]^T$ are the **corner points** of S .
- $[1, 0]^T$ lies on H_2, H_3 , which are **LI**
- $[0, 1]^T$ lies on H_1, H_4, H_5 , of which each of $\{H_1, H_4\}$,
 $\{H_1, H_5\}$, $\{H_4, H_5\}$ is **LI**

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$, $H_2 = \{[x, y]^T : y = 0\}$, $H_3 = \{[x, y]^T : x + 2y = 1\}$, $H_4 = \{[x, y]^T : 2x + 4y = 4\}$ and $H_5 = \{[x, y]^T : -x + y = 1\}$
- $[1, 0]^T, [0, 1]^T, [0, \frac{1}{2}]^T, [2, 0]^T$ are the **corner points** of S .
- $[1, 0]^T$ lies on H_2, H_3 , which are **LI**
- $[0, 1]^T$ lies on H_1, H_4, H_5 , of which each of $\{H_1, H_4\}$, $\{H_1, H_5\}$, $\{H_4, H_5\}$ is **LI**
- $[0, \frac{1}{2}]^T$ lies on H_1, H_3 , which are **LI**

- Consider the feasible region of **Example 2**

$$S = \{[x, y]^T : x \geq 0, y \geq 0, 2x + 4y \leq 4, -x + y \leq 1, x + 2y \geq 1\}$$

- The defining hyperplanes are $H_1 = \{[x, y]^T : x = 0\}$, $H_2 = \{[x, y]^T : y = 0\}$, $H_3 = \{[x, y]^T : x + 2y = 1\}$, $H_4 = \{[x, y]^T : 2x + 4y = 4\}$ and $H_5 = \{[x, y]^T : -x + y = 1\}$
- $[1, 0]^T, [0, 1]^T, [0, \frac{1}{2}]^T, [2, 0]^T$ are the **corner points** of S .
- $[1, 0]^T$ lies on H_2, H_3 , which are **LI**
- $[0, 1]^T$ lies on H_1, H_4, H_5 , of which each of $\{H_1, H_4\}$, $\{H_1, H_5\}$, $\{H_4, H_5\}$ is **LI**
- $[0, \frac{1}{2}]^T$ lies on H_1, H_3 , which are **LI**
- $[2, 0]^T$ lies on H_2, H_4 , which are **LI**

- An $\mathbf{x} \in \text{Fea}(LPP)$ which lies in **atleast one defining hyperplane** of the $\text{Fea}(LPP)$ is a **boundary point** of $\text{Fea}(LPP)$

- An $\mathbf{x} \in \text{Fea}(LPP)$ which lies in **atleast one defining hyperplane** of the $\text{Fea}(LPP)$ is a **boundary point** of $\text{Fea}(LPP)$
 $[\frac{3}{2}, 0]^T$ is a **boundary point** but **not** a **corner point** of S since it lies on only **one defining hyperplane** of S

- An $\mathbf{x} \in \text{Fea}(LPP)$ which lies in **atleast one defining hyperplane** of the $\text{Fea}(LPP)$ is a **boundary point** of $\text{Fea}(LPP)$
 $[\frac{3}{2}, 0]^T$ is a **boundary point** but **not** a **corner point** of S since it lies on only **one defining hyperplane** of S
- An $\mathbf{x} \in \text{Fea}(LPP)$ which does **not** lie on any of the **defining hyperplanes** of $\text{Fea}(LPP)$ is an **interior point** of $\text{Fea}(LPP)$

- An $\mathbf{x} \in \text{Fea}(LPP)$ which lies in **atleast one defining hyperplane** of the $\text{Fea}(LPP)$ is a **boundary point** of $\text{Fea}(LPP)$
 $[\frac{3}{2}, 0]^T$ is a **boundary point** but **not** a **corner point** of S since it lies on only **one defining hyperplane** of S
- An $\mathbf{x} \in \text{Fea}(LPP)$ which does **not** lie on any of the **defining hyperplanes** of $\text{Fea}(LPP)$ is an **interior point** of $\text{Fea}(LPP)$
 $[\frac{3}{2}, \frac{1}{8}]^T$ is an **interior point** of S since it **does not** lie on any **defining hyperplane** of S

- (Alternate definition of corner points): The **corner points** of the feasible region of a LPP **cannot** be written as a **strict convex combination** of **two distinct points** of the feasible region
- The corner points are the **extreme points** of the feasible region

- (Alternate definition of corner points): The **corner points** of the feasible region of a LPP **cannot** be written as a **strict convex combination** of **two distinct points** of the feasible region
- The corner points are the **extreme points** of the feasible region
- **Definition**: Given a nonempty **convex set**, $S \subset \mathbb{R}^n$, an $\mathbf{x} \in S$ is said to be an **extreme point** of S if $\mathbf{x}$ **cannot** be written as a **strict convex combination** of **two distinct elements** of S

- (Alternate definition of corner points): The **corner points** of the feasible region of a LPP **cannot** be written as a **strict convex combination** of **two distinct points** of the feasible region
- The corner points are the **extreme points** of the feasible region
- **Definition**: Given a nonempty **convex set**, $S \subset \mathbb{R}^n$, an $\mathbf{x} \in S$ is said to be an **extreme point** of S if $\mathbf{x}$ **cannot** be written as a **strict convex combination** of **two distinct elements** of S

That is if $\mathbf{x} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, for some $0 < \lambda < 1$ and $\mathbf{x}_1, \mathbf{x}_2 \in S$, then

- (Alternate definition of corner points): The **corner points** of the feasible region of a LPP **cannot** be written as a **strict convex combination** of **two distinct points** of the feasible region
- The corner points are the **extreme points** of the feasible region
- **Definition**: Given a nonempty **convex set**, $S \subset \mathbb{R}^n$, an $\mathbf{x} \in S$ is said to be an **extreme point** of S if $\mathbf{x}$ **cannot** be written as a **strict convex combination** of **two distinct elements** of S

That is if $\mathbf{x} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, for some $0 < \lambda < 1$ and $\mathbf{x}_1, \mathbf{x}_2 \in S$, then $\mathbf{x}_1 = \mathbf{x}_2 = \mathbf{x}$

- (Alternate definition of corner points): The **corner points** of the feasible region of a LPP **cannot** be written as a **strict convex combination** of **two distinct points** of the feasible region
- The corner points are the **extreme points** of the feasible region
- **Definition**: Given a nonempty **convex set**, $S \subset \mathbb{R}^n$, an $\mathbf{x} \in S$ is said to be an **extreme point** of S if $\mathbf{x}$ **cannot** be written as a **strict convex combination** of **two distinct elements** of S

That is if $\mathbf{x} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, for some $0 < \lambda < 1$ and $\mathbf{x}_1, \mathbf{x}_2 \in S$, then $\mathbf{x}_1 = \mathbf{x}_2 = \mathbf{x}$

- All points on the **boundary** of a **disc** are **extreme points** of the disc
- A **hyperplane**, **half space** does not have any extreme points

- **Theorem:** If $S = \text{Fea}(LPP)$ is nonempty, where
 $S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, \dots, m, -\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, \dots, n\}$ and $\mathbf{x} \in S$ then
 $\mathbf{x}$ is a **corner point** of $S \Leftrightarrow \mathbf{x}$ is an **extreme point** of S

- Theorem:** If $S = \text{Fea}(LPP)$ is nonempty, where $S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, \dots, m, -\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, \dots, n\}$ and $\mathbf{x} \in S$ then $\mathbf{x}$ is a **corner point** of $S \Leftrightarrow \mathbf{x}$ is an **extreme point** of S
- The total number of **extreme points** of the feasible region of a LPP with $(m + n)$ constraints (including the non negativity constraints) is $\leq (m + n)C_n$

- Theorem:** If $S = \text{Fea}(LPP)$ is nonempty, where $S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} \leq b_i, i = 1, \dots, m, -\mathbf{e}_j^T \mathbf{x} \leq 0, j = 1, \dots, n\}$ and $\mathbf{x} \in S$ then $\mathbf{x}$ is a **corner point** of $S \Leftrightarrow \mathbf{x}$ is an **extreme point** of S
- The total number of **extreme points** of the feasible region of a LPP with $(m + n)$ constraints (including the non negativity constraints) is $\leq (m + n)C_n$
- Exercise:** Think of a LPP $(m + n)$ constraints such that the number of **extreme points** of the $\text{Fea}(LPP)$ is equal to $(m + n)C_n$

- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly greater than $(m + n)$ and is strictly less than $(m + n)_{C_n}$

- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly greater than $(m + n)$ and is strictly less than $(m + n)_{C_n}$
- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly less than $(m + n)$

- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly greater than $(m + n)$ and is strictly less than $(m + n)_{C_n}$
- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly less than $(m + n)$
- **Question:** Does the feasible region of a LPP of the form given before (with non negativity constraints) always have an **extreme point**?

- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly greater than $(m + n)$ and is strictly less than $(m + n)_{C_n}$
- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly less than $(m + n)$
- **Question:** Does the feasible region of a LPP of the form given before (with non negativity constraints) always have an **extreme point**?

Answer: **YES**, we will see this later

- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly greater than $(m + n)$ and is strictly less than $(m + n)_{C_n}$
- **Exercise:** If possible give an example of a LPP with $(m + n)$ constraints such that the number of **extreme points** of the $Fea(LPP)$ is strictly less than $(m + n)$
- **Question:** Does the feasible region of a LPP of the form given before (with non negativity constraints) always have an **extreme point**?

Answer: **YES**, we will see this later

- **Exercise:** Think of a **convex set** defined by only one hyperplane. Will it have any **extreme point**?